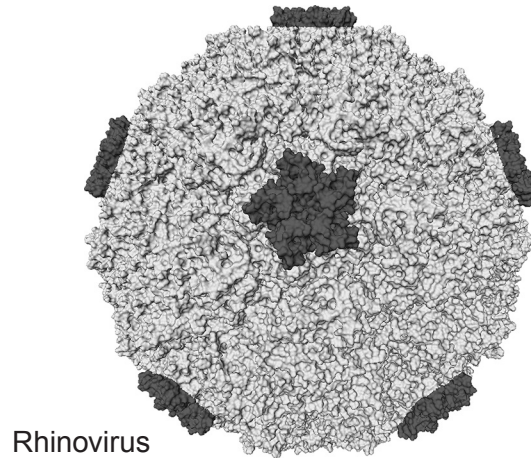


Option A: Immunology and Disease

8. The common cold is caused by a rhinovirus which infects the cells of the nasal epithelium.



Symptoms of this infection include a runny nose, sneezing, cough and sore throat. The nasal epithelium contains ciliated mucous membranes which usually act as a natural barrier to infection by trapping microbes entering in inhaled air.

- (a) (i) State **three other** natural barriers to infection in the body. [1]

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- (ii) Explain why the body is sometimes described as a host to other living organisms. [1]

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- (iii) Rhinovirus uses the metabolic pathways of a host cell to produce many copies of the virus. Explain how infection with rhinovirus could cause cell death. [1]

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- (iv) In cells infected with rhinovirus, the cell-mediated immune response involves the activation of B and T lymphocytes, and phagocytes.
Explain how phagocytes and T lymphocytes bring about the cell-mediated response against rhinovirus. [3]

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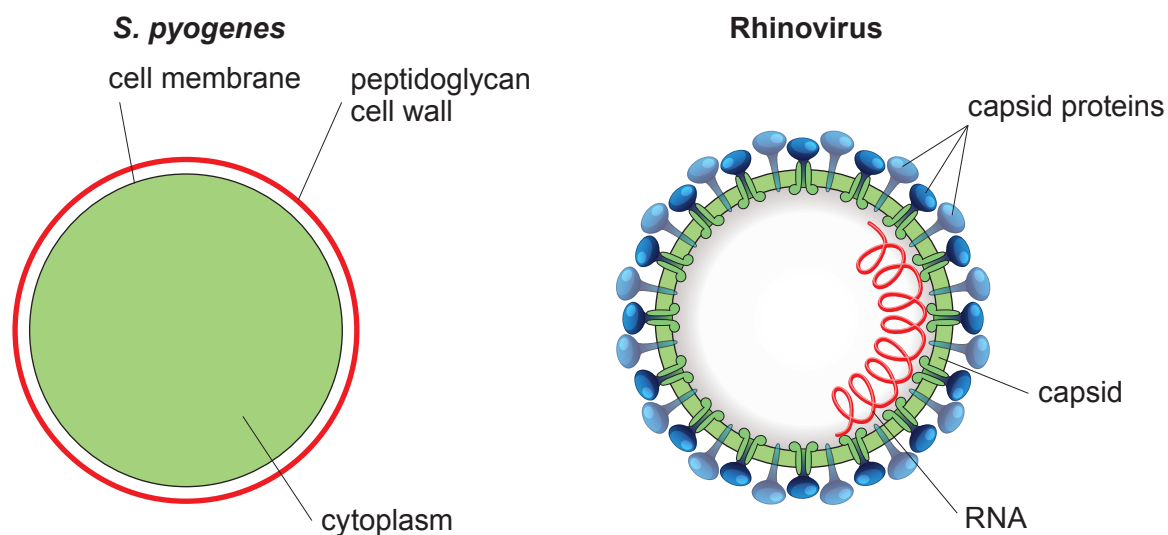
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- (b) Symptoms of a cold can also be caused by other viruses and bacterial infections. *Streptococcus pyogenes* is a species of Gram-positive bacteria which produces similar symptoms. These symptoms can be treated with antibiotics. The diagrams below show simplified structures of a coccus bacterium (such as *S. pyogenes*) and a virus (such as rhinovirus).



- (i) With reference to the diagram, explain why antibiotics can be used to treat colds caused by *S. pyogenes* but not those caused by rhinovirus. [3]

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- (ii) Patients infected with *S. pyogenes* can be treated with the antibiotic amoxycillin which has a similar mode of action to penicillin. The recommended dosage is at 25 mg kg^{-1} twice a day for 10 days. Calculate the total dosage over the 10 days for a child weighing 14.5 kg. [2]

Total dosage = mg

